Setup guide - run the Agentic RAG Platform on your machine

This guide takes you from an empty machine to a running app: a chat that answers questions from documents with citations, plus the admin pages. No knowledge of the project is needed. Commands are given for **Windows PowerShell** and for **macOS / Linux (bash)** where they differ.

At the end you will have:

What	Address
Chat + admin web app	http://localhost:8080 (Docker) or http://localhost:4200 (local development)
REST API + interactive docs (Swagger)	http://127.0.0.1:8000/docs
Health / readiness checks	http://127.0.0.1:8000/health · http://127.0.0.1:8000/ready

There are two ways to run it. Steps 1-5 are the same for both:

- **Option A - Docker** (recommended to *use* the app): one command starts API + web app.
- **Option B - local development** (to *change* the code): API and web app run in two terminals with live reload.

Contents

1. Install the tools
2. Get the code
3. Get a Gemini API key
4. Install the Python packages
5. Create the `.env` settings file
6. Check the key and index the sample documents
7. Option A - run with Docker
8. Option B - run locally for development
9. First steps in the app
10. Optional: Sign in with Google
11. Optional: LangSmith tracing
12. Command-line tool (`rag`) cheat sheet
13. Tests and quality checks
14. Where data is stored, and how to reset
15. Troubleshooting

---

1. Install the tools

Tool	Needed for	Version	Get it
Git	getting the code	any	https://git-scm.com/downloads
uv	Python packages and the `rag` command (installs **Python 3.12** by itself)	0.13 or newer	see below
Docker Desktop	Option A	recent	https://www.docker.com/products/docker-desktop/
Node.js (includes npm)	Option B (the web app in development)	**24**	https://nodejs.org/

You do **not** need to install Python yourself: `uv` downloads the right version (3.12) on first use. (.NET analogy: uv is like the `dotnet` CLI + NuGet in one tool.)

Install uv

Windows PowerShell:

```
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

macOS / Linux:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Close and reopen the terminal, then check everything:

```
git --version
uv --version
docker --version      # Option A
node --version        # Option B: should print v24.x
```

Expected output (your version numbers may be newer):

```
git version 2.49.0.windows.1
uv 0.13.0 (4ecd0eff2 2026-10-09 x86_64-pc-windows-msvc)
Docker version 27.4.0, build bde2b89
v24.20.0
```

---

2. Get the code

```
git clone https://github.com/ratheesh-5/agentic-rag-platform.git
cd agentic-rag-platform
```

All later commands are run **from this folder** (the project root) unless a step says otherwise.

> **Windows: use a short folder path**, e.g. `C:\projects\agentic-rag-platform`. Some Python package files sit deep inside `.venv`, and Windows refuses paths longer than 260 characters unless "long paths" are enabled. In a very deep folder the install looks fine but the first command fails with "*ImportError: ... No module named ...*" (seen while testing this guide).

---

3. Get a Gemini API key

The app uses Google **Gemini** for answers and for embeddings (turning text into vectors for search). The free tier is enough to try everything.

1. Open https://aistudio.google.com/apikey and sign in with a Google account.
2. Click **Create API key** and copy it. Treat it like a password: never commit it or paste it into chats or tickets.

Free-tier limits (per model): about **15 chat requests per minute and 500 per day**, and **100 embedded texts per minute**. The app throttles itself to stay inside these limits.

---

4. Install the Python packages

```
uv sync
```

Expected output (first run; takes about a minute when packages must be downloaded):

```
Using CPython 3.12.15
Creating virtual environment at: .venv
Resolved 123 packages in 1ms
   Building rag-platform @ file:///C:/projects/agentic-rag-platform
      Built rag-platform @ file:///C:/projects/agentic-rag-platform
Prepared 1 package in 2.01s
Installed 122 packages in 4.32s
 + aiohappyeyeballs==2.7.1
 ... (one line per package)
```

This creates a virtual environment in `.venv/` (like a project-local package folder) with every package from `uv.lock`, and installs the `rag` command. Run commands through uv, e.g. `uv run rag --help` - no "activate" step needed.

---

5. Create the `.env` settings file

Settings and secrets live in a `.env` file in the project root. It is **git-ignored**; the tracked template is `.env.example`.

Windows PowerShell:

```
Copy-Item .env.example .env
```

macOS / Linux:

```
cp .env.example .env
```

Open `.env` in an editor and fill in the two required values:

```
GEMINI_API_KEY=your-gemini-key-from-step-3
API_KEYS=a-long-random-admin-key
```

`API_KEYS` are the **admin keys** for the web app's "Admin sign in" and for scripts (comma- separated if you want several). Make a strong random one with:

```
uv run python -c "import secrets; print(secrets.token_urlsafe(32))"
```

Optional settings (uncomment the line in `.env` to change a default):

Setting	Default	What it does
`CHAT_MODEL`	`gemini-3.5-flash-lite`	Default answer model. If it is overloaded (errors 503/504), try e.g. `gemini-3.1-flash-lite`: each model has its own quota. Admin › Settings › Models can choose another model (and a backup model) without editing `.env`
`EMBEDDING_MODEL`	`gemini-embedding-001`	**Don't change** unless you re-index every document
`GROQ_API_KEY`	empty	Optional second answer model (Groq, free key): takes over when Gemini is out of quota or overloaded; chosen in Admin › Settings › Answer model. Search keeps using Gemini
`GROQ_MODEL`, `GROQ_REQUESTS_PER_MINUTE`	`openai/gpt-oss-120b`, 5	Groq model and client-side limit (free tier: ~8,000 tokens per minute)
`GUEST_QUESTIONS_PER_HOUR`	20	Questions per hour for visitors who don't sign in (per IP); 0 = guests off
`GOOGLE_CLIENT_ID`, `ADMIN_EMAILS`	empty	Sign in with Google - see section 10
`USER_QUESTIONS_PER_HOUR`	50	Per signed-in Google user
`SESSION_DAYS`	7	How long a Google sign-in lasts
`USER_MAX_COLLECTIONS`, `USER_MAX_DOCUMENTS`, `USER_MAX_PASSAGES`	5, 20, 500	What one signed-in user may store in private collections
`CHAT_RETENTION_DAYS`, `USAGE_RETENTION_DAYS`	90	Chats without activity / usage-log rows older than this are deleted daily; 0 = keep
`ONLINE_EVAL_SAMPLE_RATE`	0.2	Share of chat answers scored in the background (1 in 5; uses extra AI calls); 0 = off
`MAX_UPLOAD_MB`	10	Largest document upload
`CORS_ORIGINS`	`http://localhost:4200`	Web apps on another address that may call the API (Option B)
`LANGSMITH_*`	off	Tracing - see section 11

Search settings (chunk size, number of passages, hybrid search, similarity threshold, answer mode, answer model) are **not** in `.env`: they live in `rag_config.json` and are changed from the app (Admin › Settings).

---

6. Check the key and index the sample documents

```
uv run rag check
```

Makes one chat call per chat provider (Gemini, plus Groq if `GROQ_API_KEY` is set) and one embedding call. Expected output (the chat model is your `CHAT_MODEL`):

```
chat model (gemini): gemini-3.1-flash-lite -> 'OK'
chat model (groq): openai/gpt-oss-120b -> 'OK'
embedding model: gemini-embedding-001 -> 3072 dims
OK
```

If it fails, the message says why (wrong key, quota, model name).

Now index the six demo documents in `data/sample_corpus/` (a fictional company handbook, "Lumora Labs"):

```
uv run rag ingest
```

The first run embeds about 80 passages; it took **16 seconds** when this guide was tested (the free tier allows 100 texts per minute, so a bigger corpus waits in between). Expected output:

```
docs:   6 added, 0 updated, 0 unchanged, 0 deleted
chunks: 78 added, 0 removed
embeddings: 78 new, 0 from cache
store: 6 docs, 78 chunks  (data\index)
```

Later runs only embed new or changed text, so re-running it is quick and free (`0 new, 78 from cache`).

Try one question from the command line:

```
uv run rag ask "What does error code LUM-2001 mean?"
```

Expected output - the answer with a citation and its source passage (character range and similarity score):

```
Error code LUM-2001 means that a model gateway timeout has occurred, specifically when a
request exceeds 30 seconds [2].

Sources:
  [2] data/sample_corpus/llmops_incident_runbook.md [1498:2290]  score 0.745
```

This is the whole RAG pipeline on the command line: search → answer → citation check. (The wording of the answer can differ slightly from run to run.)

> Prefer not to install uv at all? With Docker only, skip this step and upload documents later in **Admin › Documents** in the web app.

---

7. Option A - run with Docker

Start Docker Desktop first and wait until it says it is running. Then, in the project root:

```
docker compose up --build -d
```

- `--build` builds the two images (API and web app); the first build takes a few minutes.
- `-d` runs them in the background.

If port 8080 is already used by another program, choose another port:

Windows PowerShell:

```
$env:UI_PORT = "8081"; docker compose up --build -d
```

macOS / Linux:

```
UI_PORT=8081 docker compose up --build -d
```

Then open **http://localhost:8080** (or your `UI_PORT`). The API docs are at http://127.0.0.1:8000/docs (only reachable from your own machine).

Check that both containers are healthy:

```
docker compose ps
```

Expected output (shortened; here with `UI_PORT=8081`) - both say `(healthy)` after about half a minute:

```
NAME                SERVICE   STATUS                  PORTS
rag-platform-api-1  api       Up 12 hours (healthy)   127.0.0.1:8000->8000/tcp
rag-platform-ui-1   ui        Up 12 hours (healthy)   0.0.0.0:8081->8080/tcp
```

http://127.0.0.1:8000/ready shows the details (settings file, database, search index):

```
{"status":"ready","checks":{"model_key":"ok","config":"ok","database":"ok",
 "search_index":"ok (94 passages)"}}
```

Everyday Docker commands

Task	Command
Stop	`docker compose down`
Start again (no code change)	`docker compose up -d`
After pulling new code	`docker compose up --build -d`
After editing `.env`	`docker compose up -d --force-recreate api`
API log (JSON lines, one per request / AI call / answer)	`docker compose logs api --tail 20`
Follow the log live (Ctrl+C to stop)	`docker compose logs api -f`

Remember the `UI_PORT` setting each time you run `up` in a new terminal if you use one.

What the API log looks like - one question, all lines tied together by the same `request_id`: an embedding call for the search, the answer call, the outcome and the HTTP request. The question text is never logged. (Lines shortened.)

```
{"msg": "model call", "request_id": "28757c317070", "model": "gemini-embedding-001",
 "purpose": "embed", "texts": 1, "latency_ms": 696}
{"msg": "model call", "request_id": "28757c317070", "model": "gemini-3.1-flash-lite",
 "purpose": "answer", "input_tokens": 1133, "output_tokens": 94, "latency_ms": 1854}
{"msg": "answer", "request_id": "28757c317070", "route": "fast", "answerable": true,
 "citations": 1, "passages": 5}
{"msg": "request", "request_id": "28757c317070", "method": "POST", "path": "/chat",
 "auth": "guest-id", "status": 200, "ms": 3100}
```


Your documents, the index, chats and `rag_config.json` stay in the project folder (`data/index/` , `data/uploads/`), so they survive restarts and rebuilds.

---

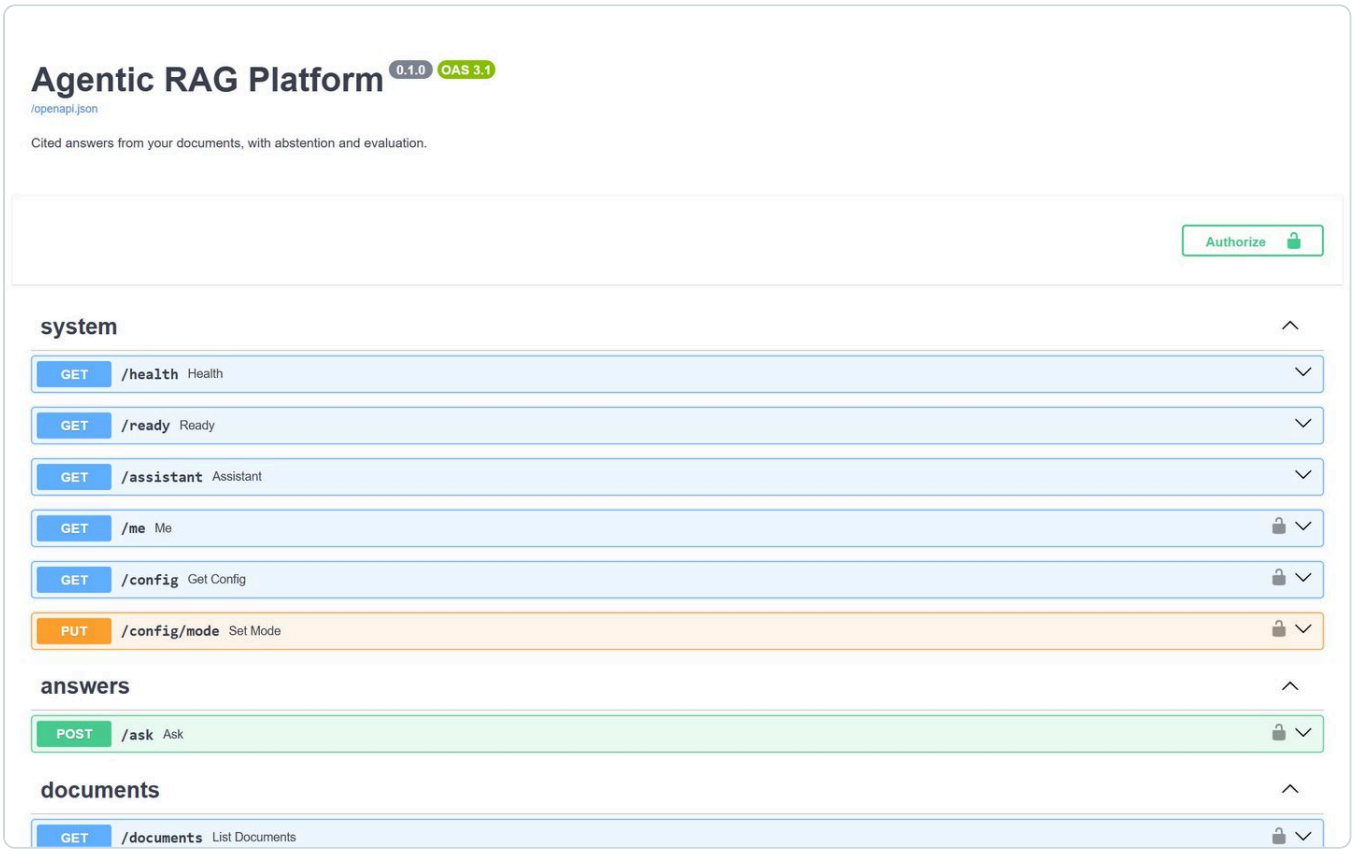
8. Option B - run locally for development

Use two terminals, both starting in the project root.

Terminal 1 - the API (restarts itself when you edit Python code):

```
uv run rag serve --reload
```

API at http://127.0.0.1:8000, docs at http://127.0.0.1:8000/docs - every endpoint can be tried there (**Authorize** takes an admin key):



Terminal 2 - the web app:

```
cd ui
npm install          # first time only (and after package.json changes)
npm start
```

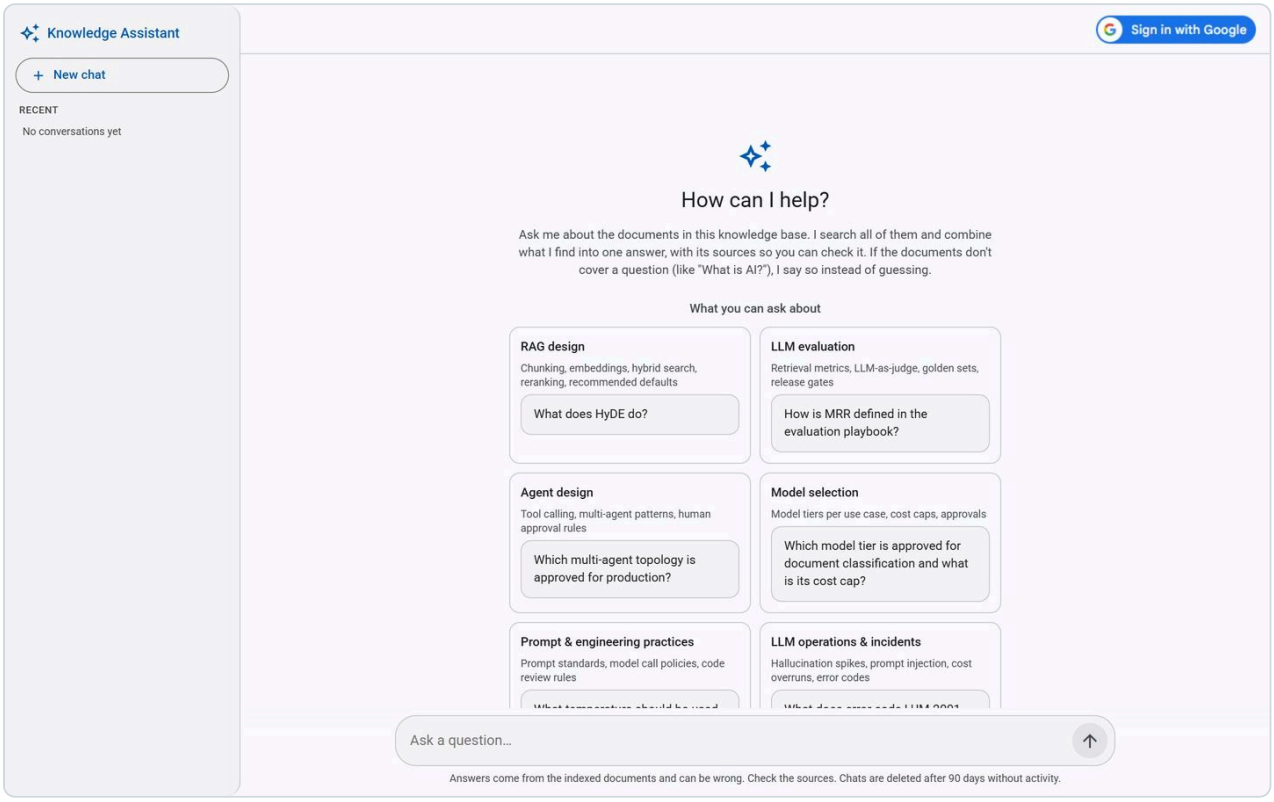
Open **http://localhost:4200** (use `localhost`, not `127.0.0.1`: the API allows the web app only from the addresses in `CORS_ORIGINS`, and Google sign-in checks the address exactly). The page reloads by itself when you edit the Angular code.

Stop either one with **Ctrl+C**.

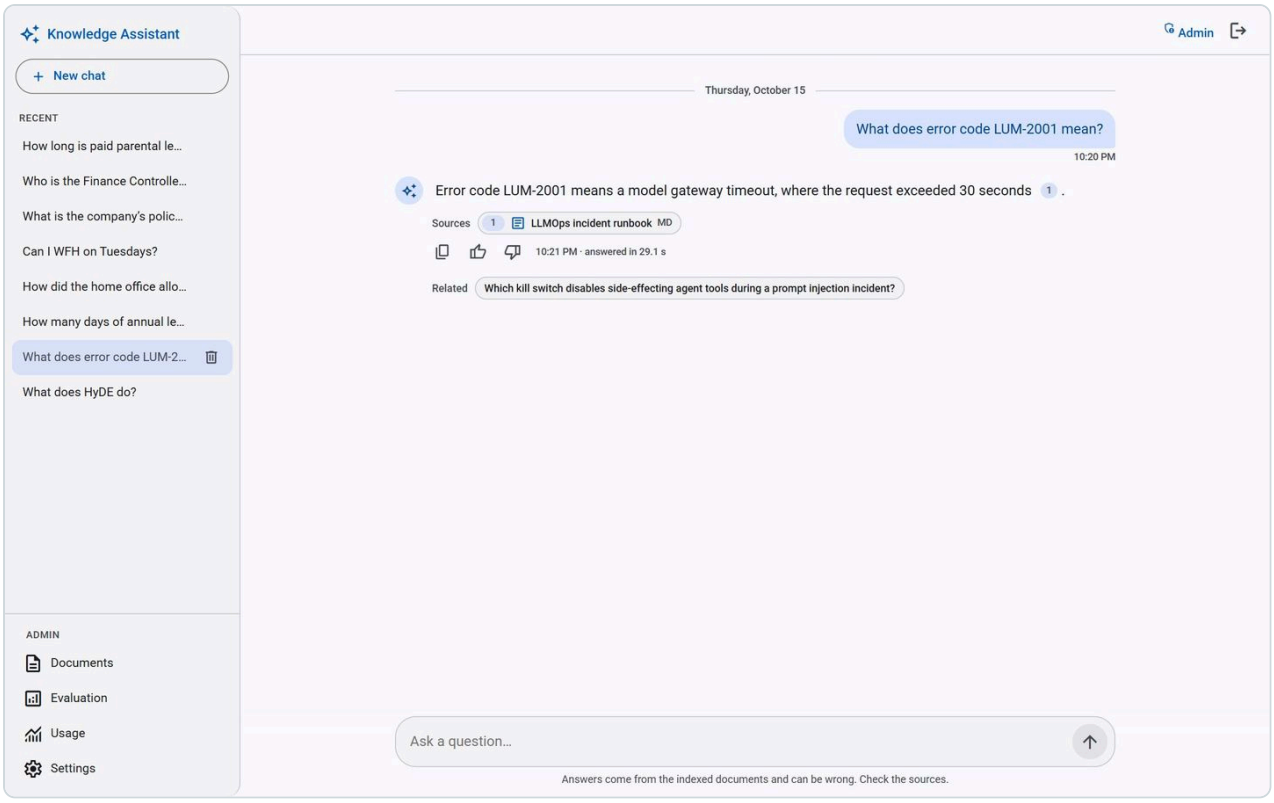
---

9. First steps in the app

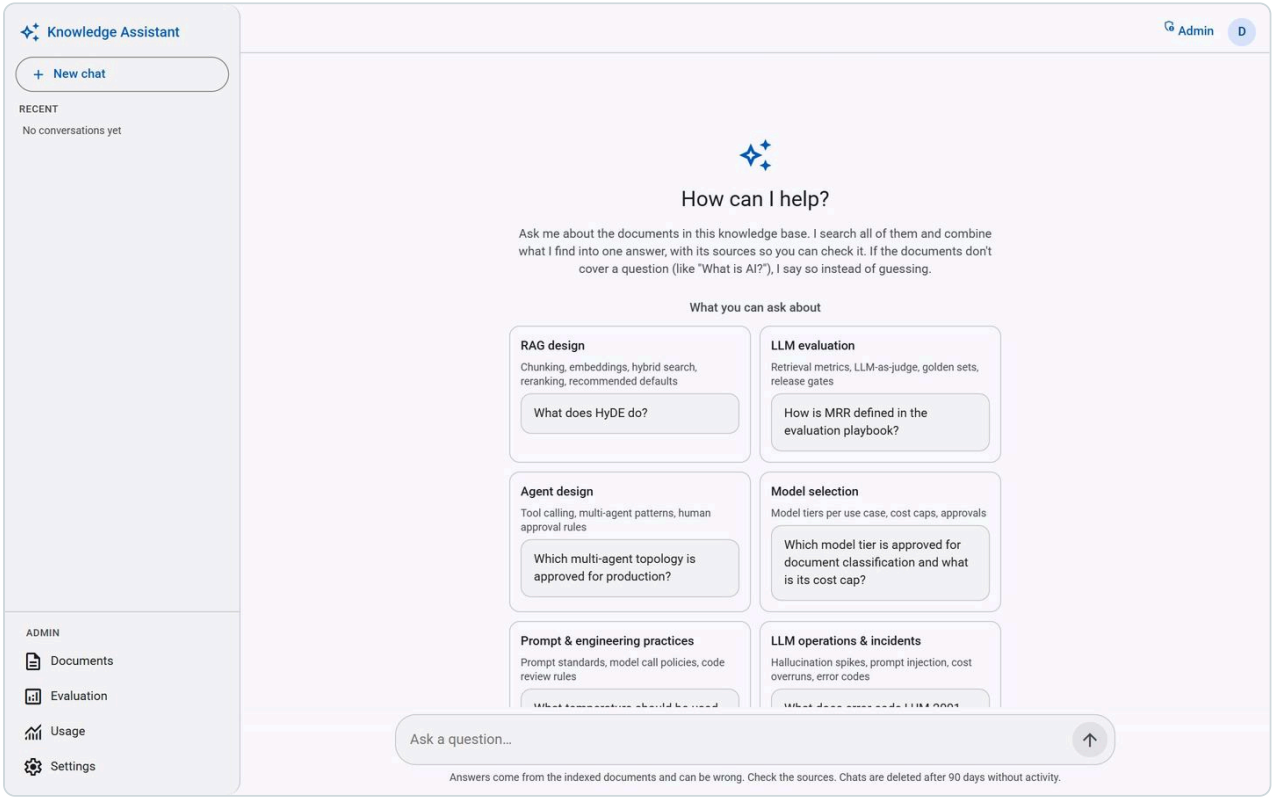
1. **Chat as a guest** - no sign-in needed. The welcome screen shows topics and example questions; click one, or ask e.g. "*What is the default chunk size?*". Off-topic questions get "*I couldn't find this in the documents*" instead of a guess.



   Answers show numbered citations and source chips (click one to see the passage):



2. **Become admin** - click **Admin sign in** (top right) and paste one of your `API_KEYS`. If Google sign-in is configured, sign in with an email listed in `ADMIN_EMAILS` instead.
3. **Admin pages** (left sidebar, only for admins):



   - **Documents** - collections at the top (**New collection**, rename, delete); upload PDF, Word (.docx), Markdown or text files (drag and drop) into the selected collection; they can be asked about from the next question.

- **Evaluation** - *Run quality check* measures search quality on the 50-question test set in about a second, with no AI calls; *Live quality* shows how real chat answers score.
- **Usage** - AI calls, tokens, latency and errors.
- **Settings** - the active configuration, the answer mode (Fast / Agentic / Auto), and *Find better settings* (the multi-agent optimizer; makes a few AI calls).

---

10. Optional: Sign in with Google

Lets people sign in with their Google account: their chats follow them to every device, and admins are named people (by email) instead of a shared key. The setup is done **once by whoever runs the app**; users only click the button. Nothing here needs a credit card.

1. **Project:** open https://console.cloud.google.com/projectcreate, give it a name (e.g. `knowledge-assistant`), keep **Location: No organization**, click **Create**, then select the project in the project picker at the top.
2. **Sign-in app:** open https://console.cloud.google.com/auth/overview → **Get started**: app name (e.g. *Knowledge Assistant*), support email, **Audience: External**, contact email, accept, **Create**.
3. **Test users:** https://console.cloud.google.com/auth/audience → **Test users** → **Add users**: the Google accounts allowed to sign in while the app is in testing (up to 100).
4. **Client ID:** https://console.cloud.google.com/auth/clients → **Create client** → **Web application**. Under **Authorized JavaScript origins** add:

   - `http://localhost`
   - `http://localhost:4200` (Option B)
   - `http://localhost:8080` and/or your `UI_PORT`, e.g. `http://localhost:8081` (Option A)

   Leave redirect URIs empty, click **Create** and copy the **Client ID** (ends with `.apps.googleusercontent.com`). Ignore the client secret - this app doesn't use one.
5. **`.env`:**

   ```
   GOOGLE_CLIENT_ID=1234567890-abc.apps.googleusercontent.com
   ADMIN_EMAILS=you@gmail.com
   ```
6. Restart the API (Docker: `docker compose up -d --force-recreate api`; Option B: stop and start `rag serve`), reload the page: the toolbar now shows **Sign in with Google**.

New Google clients can take a few minutes to start working. With an empty `GOOGLE_CLIENT_ID` the app simply shows the "Admin sign in" (key) button instead.

---

11. Optional: LangSmith tracing

Shows every request as a tree of steps (prompts, tokens, timings) in the LangSmith web app.

1. Create a free account at https://smith.langchain.com and an API key (Settings → API Keys).
2. In `.env`:

```
LANGSMITH_TRACING=true
LANGSMITH_API_KEY=your-langsmith-key
LANGSMITH_PROJECT=rag-platform
```

3. Restart the API. The API is traced automatically; command-line runs only with `uv run rag --trace-langsmith <command>` (so test runs don't use up the free monthly allowance).

---

12. Command-line tool (`rag`) cheat sheet

Run `uv run rag --help` or `uv run rag <command> --help` for every option.

Command	What it does	AI calls
`uv run rag check`	Test the Gemini key	2
`uv run rag inspect`	Show how documents are split into passages	none
`uv run rag ingest [folder-or-file] [--collection NAME]`	Index documents (default `data/sample_corpus`, into the demo collection; a new NAME creates the collection); only new/changed text is embedded	embeddings for new text
`uv run rag delete <path>`	Remove a document from the index	none
`uv run rag ask "question"`	Answer one question; add `--collection NAME`, `--mode agentic`, `--trace`, `--show-context`	1 (fast) / ~4 (agentic)
`uv run rag eval run --retrieval-only`	Search-quality metrics on the test set	none
`uv run rag eval run --limit 5`	Answers + RAGAS scores on 5 test questions	~12 per question
`uv run rag eval list` / `eval compare 1 2`	List / compare evaluation runs	none
`uv run rag optimize`	Agents propose better settings; you approve or reject	a few
`uv run rag serve`	Start the REST API (`--reload` for development)	-
`uv run rag gate`	The CI quality gate, locally (no API key needed)	none

---

13. Tests and quality checks

None of these call Gemini: the tests use fake models.

```
uv run pytest                          # backend tests
uv run ruff check .                    # Python lint
uv run ruff format --check .           # Python formatting
uv run rag gate                        # retrieval quality gate (Recall@5, MRR)
```

Expected output of `pytest` (last line) and `rag gate`:

```
193 passed in 58.89s
```

```
  PASS  Recall@5            1.000 >= 0.85
  PASS  MRR                 0.881 >= 0.7
Quality gate passed.
```

Web app (in `ui/`):

```
cd ui
npm test -- --watch=false                      # UI tests (Vitest)
npx prettier --check "src/**/*.{ts,html,scss}" # formatting
npm run build                                  # production build
```

GitHub Actions runs all of these (plus a dependency audit and the Docker builds) on every push; see `.github/workflows/ci.yml`.

---

14. Where data is stored, and how to reset

Path	Contents	In Git?
`.env`	Your keys and settings	**no** (secret)
`rag_config.json`	Active search settings and answer mode	yes
`assistant.json`	Topics and example questions shown in the chat	yes
`data/sample_corpus/`	The six demo documents	yes
`data/eval/`	Test set (`golden.jsonl`), release gates, CI snapshot	yes
`data/index/rag.db`	SQLite: documents, passages, embedding cache, chats, users, evaluation runs, usage log	no
`data/index/faiss.index`	Vector search index of the demo collection (rebuilt automatically from the database)	no
`data/index/collections/`	One vector search index per other collection	no
`data/uploads/`	Documents uploaded in the web app	no

Start over completely: stop the app, delete the `data/index/` folder, then run `uv run rag ingest` again. Note that this also deletes chats, users and evaluation history. (The FAISS file alone can be deleted any time: it is rebuilt from the database without AI calls.)

---

15. Troubleshooting

Problem	Cause and fix
`uv` or `node` "not recognized" after installing	Close and reopen the terminal (the PATH changed)
Right after installing, `rag ...` fails with *"ImportError ... No module named ..."* (Windows)	The project folder path is too long (260-character limit): clone into a short folder such as `C:\projects\` and run `uv sync` again
`rag check` fails with an auth/permission error	Wrong `GEMINI_API_KEY` in `.env`, or a space/quote around it
Answers fail with *"Model quota exceeded"* (HTTP 429)	Free-tier limit reached: wait a minute (or until the next day for the daily limit), set another `CHAT_MODEL`, choose a **backup model** in Admin › Settings › Models (with Automatic fallback on), or add `GROQ_API_KEY` so Groq takes over
"The AI service is slow or overloaded" (HTTP 503)	Google's model is busy: retry, choose another model or a backup model in Admin › Settings › Models (no restart), or switch `CHAT_MODEL` (e.g. `gemini-3.1-flash-lite`) and restart the API
Every question says *"I couldn't find this"*	Nothing is indexed: run `uv run rag ingest` or upload documents; check `/ready` shows passages
Port 8080 already in use	Use `UI_PORT` (see section 7)
`UI_PORT=8081 docker compose ...` fails in PowerShell	That is bash syntax; use `$env:UI_PORT = "8081"; docker compose ...`
Docker: `env file .env not found`	Create `.env` first (step 5)
`docker compose ps` shows `unhealthy`	Open http://127.0.0.1:8000/ready - it names the failing check; see `docker compose logs api`
Docker Desktop won't start on Windows (*"removing stale socket ... userAnalyticsOtlpHttp.sock"*)	Quit Docker Desktop, rename the folder `%LOCALAPPDATA%\Docker\run`, start Docker Desktop again. Don't use "Reset to factory defaults" (it deletes all images and volumes)
Web app (Option B) shows *"can't be reached"*	The API isn't running: start `uv run rag serve` in terminal 1
Browser error about CORS (Option B)	Open the app as `http://localhost:4200`, or add your address to `CORS_ORIGINS`
Google sign-in: *"origin not allowed"* / button missing	The page address must be listed exactly in the client's JavaScript origins (use `localhost`, not `127.0.0.1`); wait a few minutes after creating the client; the account must be a test user
`rag ingest` fails on a file starting with `~$`	A Word lock file (Word still has a document open): close Word or delete that file
Admin pages say *"Invalid or missing API key"*	The pasted key is not in `API_KEYS`, or the API wasn't restarted after editing `.env`

More detail: README.md (features, API endpoints) · design.md (architecture and flow diagrams).